



CALVIN JUDE DSOUZA



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EDUCATION

NMAM Institute of Technology (NMAMIT) | 9.26(currently)
B.Tech in Cybersecurity

Karkala, India
Aug. 2024 – May. 2028

Indian High School, Dubai | 8.91
Higher Secondary Education

Dubai, UAE
Apr. 2010 – Apr. 2022

TECHNICAL SKILLS

Languages & Frameworks: Python, JavaScript, TypeScript, SQL, HTML/CSS, Flask, Fast API, Node.js, REST API Development
AI & Machine Learning: PyTorch, YOLOv8, Computer Vision, LLM Integration, Multi-Modal Fusion Models, Pattern Recognition
Developer Tools: Linux (Kali/Ubuntu), Git, Docker, Bash Scripting, Chrome Extension Development
Cybersecurity: Digital Forensics, OSINT, Network Security, Cryptography, Steganography, Nmap, Wireshark, Gobuster

INTERNSHIPS & EXPERIENCE

NMAM Institute of Technology (NMAMIT) | Cybersecurity Developer Intern & Team Lead (Vision) | [Jan. 2025– May. 2025]

- Directed a 6-member development team under faculty mentorship to engineer "VisionEX," a web-based cybersecurity suite providing enterprise-grade digital certification tools.
- Implemented robust system encryption protocols utilizing the Diffie-Hellman key exchange to establish secure, authenticated communication channels.
- Coordinated project timelines and task delegation while integrating Minimax algorithms to optimize the platform's core cryptographic decision-making processes.

PROJECTS

Lucent | AI Multi-Parameter | Deepfake Detection Python, PyTorch, Multi-Modal Fusion, OpenCV

- Designing a deep learning framework to detect synthetic media by fusing spatial and frequency-domain analysis.
- Prototyping a multi-parameter detection engine to identify synthetic inconsistencies, utilizing Celery and Redis for asynchronous high-compute video processing.

IMS Enterprise | Full-Stack Inventory Architecture JavaScript (Node.js), PostgreSQL, React, Express

- Engineered real-time analytical dashboards that replaced legacy manual tracking, optimizing inbound/outbound logistics for high-volume operations.
- Architected an independent inventory system prototype for a **20M+ AED** food supply chain, successfully managing a simulated high-volume database of **1,000+ SKU** entries.

MassEd.ex | Computer Vision & Crowd Intelligence Python, YOLOv8, OpenCV, DeepSORT

- Engineered a high-precision crowd monitoring system using YOLOv8, successfully tracking up to 50 concurrent objects at 30 FPS during simulated stress tests using public safety video datasets.
- Developed automated "Danger-Zone" logic to trigger safety alerts, accurately identifying 4 distinct behavioral anomaly patterns in test footage.
- Applied AI vision models for behavioral analytics, focusing on pattern recognition in public safety environments

NetRecon | Deep-Analysis LAN & Topology Scanner | C, Unix Sockets, Bash, TCP/IP, awk

- Engineered a deep-analysis reconnaissance tool utilizing Unix raw sockets and shell scripting, autonomously mapping a /24 local subnet (up to 254 hosts) in under 12 seconds.
- Implemented custom ARP and ICMP traffic analysis, successfully identifying 100% of injected rogue devices during simulated local network tests.

COMPETITIONS & CERTIFICATIONS

- IISc Bangalore Cybersecurity Workshop:** Completed intensive 2-day technical training on advanced security frameworks and threat vectors.
- CYSECK NITK Workshop & CTF:** Engaged in a 2-day cybersecurity awareness training and applied tactical skills to secure **14th place out of 60+ teams** in the concluding Capture The Flag competition.
- Certifications in Progress:** Actively preparing for CompTIA Security+, CHFI, and eJPT to formalize expertise in network security, incident response, and post-breach analysis.
- Capture The Flag (CTF) Competitor:** Competed in 10+ CTF events, specializing in digital forensics, reverse engineering, web vulnerabilities, and OSINT. Secured 7th place at *Code Intrusion* and 14th place at *Enyugma CTF* out of a highly competitive field of 200+ participants.
- National-Level Hackathons (SIH, Innovex, Protothon):** Actively participated in and developed rapid solutions for multiple national hackathons. Received special commendation from judges at **Innovex** for exceptional strategic thinking, innovative problem-solving, and high-energy execution.

LEADERSHIP & EXTRACURRICULARS

- Junior Branch Representative | NMAMIT Cybersecurity Dept. (Jan 2026 – 2027):** Orchestrated branch-level festival logistics and task delegation for cross-functional teams of 20+ members. Managed budget allocation and on-ground execution for high-attendance campus events.
- Class Representative | NMAMIT (Aug 2025 – 2026):** Served as the primary strategic liaison between 70+ students and faculty to streamline academic coordination.
- Core Member – PROTON | College Association (Aug 2025 – 2026):**
 - Coordinated large-scale college event logistics, focusing on backend resource planning and real-time execution. Facilitated and co-organized multiple collegiate Capture The Flag (CTF) competitions, managing operational and technical logistics for participants.
 - Conceptualized, planned, and hosted highly engaging campus events, including "Game of Conquest" (a 3-tier strategy game) and "Fish Tank" (a creative entrepreneurial pitch competition).